

AI in Consumer Behavior, Marketing & Digital Ecosystems – Dr. Vishnu Prasad

I have always been someone who asks "why does this work the way it does?" — and that curiosity is honestly what brought me to this research theme. Every time I open an app and see a product recommendation that feels oddly accurate, or notice how my social media feed seems to know what I am thinking before I do, I find myself wanting to understand the system behind it. Not just technically, but behaviorally. Why does it work on people? How does it influence what we buy, what we believe, or what we pay attention to? These questions genuinely excite me, and they are the reason I chose AI in Consumer Behavior, Marketing and Digital Ecosystems as my area of interest.

I am currently in my third year of a B. Tech in Computer Science Engineering, specializing in Artificial Intelligence and Machine Learning. My coursework has given me a working understanding of machine learning algorithms, data structures, and statistical thinking. I have been steadily building hands-on skills in Python, and I work regularly with libraries like NumPy, Pandas, and Scikit-learn for data handling and analysis. But what my academic journey has also taught me is that knowing how to build a model is only half the picture. The more interesting question is what that model is actually doing to real people in real situations — and consumer behavior is one of the richest places to explore that.

What drew me specifically to this theme is how interdisciplinary it is. It is not purely a machine learning problem, and it is not purely a psychology or marketing problem either. It sits right in the middle, and that is what makes it intellectually demanding in a way I genuinely enjoy. Understanding consumer behavior through AI means you have to think about data science, human motivation, digital design, and ethics all at once. That kind of layered thinking is something I actively look for in the problems I choose to work on.

One experience that shaped my thinking in this direction was a project I worked on involving an AI healthcare assistant. I spent a lot of time exploring how AI agents process user inputs, retain context, and explain their reasoning in ways that feel trustworthy. Healthcare and consumer behavior are different domains, but the underlying dynamic is quite similar — in both cases, you are building a system that needs to understand human needs, respond meaningfully, and maintain the user's trust over time. That project pushed me to think carefully about the relationship between AI decision-making and human behavior, which I

now see as directly relevant to how recommendation systems and personalization engines function in commercial digital ecosystems.

Outside of coursework and projects, I spend a fair amount of time reading about how major platforms approach personalization, how e-commerce companies use behavioral data to improve conversion, and what the ongoing debates around algorithmic transparency and data ethics actually involve. I do not just find this interesting from a technical perspective — I find it important. AI systems that influence consumer choices at scale carry real responsibility, and I want to work in a research environment where that responsibility is taken seriously.

What I am looking for in this internship is the opportunity to develop more rigorous research habits — reading academic literature critically, working with behavioral datasets in a structured way, and learning how to frame research questions that are both technically grounded and practically meaningful. I know I am still building expertise in several areas, and I want to be straightforward about that. But I am a fast learner who is genuinely motivated by this subject, and I am prepared to put in the work that serious research demands.

This theme connects everything I care about in AI — the technical depth, the human dimension, and the broader question of what these systems mean for society. That is why I chose it, and that is why I am enthusiastic about contributing to it.